ANSWER KEY & MARKING SCHEME SET A

Chapter 13: **Earth as a System: Energy, Matter and Life** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 1 kWh	energy = power × area × time
2	**(b)** 2800 J	1400 × area
3	**(d)** 100 × 10^{-9} m	1 nm = 10^{-9} m
4	**(c)** solar radiation	the Sun
5	**(a)** infrared	heat
6	**(d)** the Sun's rays are spread over a larger area	slanting rays
7	**(c)** carbon dioxide, methane and water vapour	trap heat
8	**(b)** Montreal Protocol	ozone recovery
9	**(c)** fresh snow	high albedo
10	**(c)** condensation	cloud formation

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Geosphere, hydrosphere, atmosphere and biosphere (cryosphere may be mentioned) (2).	2
12	Warmer sea water increases evaporation and changes monsoon rainfall (2).	2
13	Energy from the Sun in the form of electromagnetic waves (1); drives winds, the water cycle and photosynthesis (1).	1+1
14	Solar constant: energy at the top of the atmosphere (about 1.4 kW m^{-2}); insolation: radiation reaching the surface (1+1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Warming melts glaciers (1); sea level rises, flooding coasts (hydrosphere) (1); affects ecosystems in the biosphere (1).	1+1+1
16	Radio, microwave, infrared, visible, ultraviolet, X-rays, gamma rays (2); all travel at 3 × 10^8 m s^{-1} in vacuum (1).	2+1
17	Harmful: damages eyes and skin, may cause cancer (1.5); useful: kills germs in water purifiers (1.5).	1½+1½
	OR: Power = 5 kW (1); energy = 5 × 2 = 10 kWh (1) = 3.6 × 10^7 J (1).	1+1+1
18	The atmosphere absorbs, scatters and reflects part of the radiation (3).	3
	OR: It lies in the tropical and sub-tropical regions and gets abundant sunshine all year (3).	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Geosphere, hydrosphere, atmosphere, biosphere (2). (ii) Warmer Arabian Sea alters the monsoon, affecting rivers and crops (2). (iii) Stores freshwater and regulates sea level (1).	2+2+1
	OR: (i) Cryosphere/hydrosphere, biosphere (1). (ii) Less snow → less melt water → shrinking lake → less grass (2). (iii) Less food for grazers (1). (iv) Water conservation and afforestation (1).	1+2+1+1
20	(i) EM energy from the Sun (1). (ii) Gamma to radio; same speed in vacuum (2). (iii) UV absorbed by ozone; visible gives light for photosynthesis; infrared warms the Earth (2).	1+2+2
	OR: (i) 4 kW (1). (ii) 20 kWh (1). (iii) 20 × 3.6 × 10^6 = 7.2 × 10^7 J (2). (iv) The solar constant (1.4) is at the top of the atmosphere, 1 kW m^{-2} at the surface (1).	1+1+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET B

Chapter 13: **Earth as a System: Energy, Matter and Life** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 6 kWh	energy = power × area × time
2	**(a)** 7000 J	1400 × area
3	**(b)** 400 × 10^{-9} m	1 nm = 10^{-9} m
4	**(d)** can travel through vacuum	unlike sound waves
5	**(b)** insolation	incoming solar radiation
6	**(c)** nitrogen	78%
7	**(c)** keeps the Earth warm enough for life	natural but enhanced by humans
8	**(d)** uneven heating of the Earth's surface	pressure differences
9	**(c)** fraction of incoming solar radiation reflected by a surface	reflectivity
10	**(b)** oceans	~71%

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	1 kW × 1 h = 1 kWh = 3.6 × 10^{6} J (2).	2
12	The Sun's rays slant and spread over a larger area (2).	2
13	Greenhouse gases trap heat re-radiated from the Earth, keeping it warm (2).	2
14	CO_2 from burning fossil fuels; methane from livestock or paddy fields (1+1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	The spherical shape gives different angles of rays (1.5); the tilt changes the angle and length of days during the year (1.5).	1½+1½
16	Troposphere: weather; stratosphere: ozone; mesosphere: meteors burn; thermosphere/exosphere: thin (3).	3
17	Absorbs harmful UV; keeps the Earth warm by the greenhouse effect (1.5+1.5).	1½+1½
	OR: Venus has a thick atmosphere of CO_2 with a strong greenhouse effect (3).	3
18	CFCs release chlorine that destroys ozone (1.5); the Montreal Protocol banned CFCs, so the layer is slowly recovering (1.5).	1½+1½
	OR: Day: land heats faster so air rises and cool sea air moves in (1.5); night: reverse (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Shape, tilt, latitude and surface type (2). (ii) Pressure differences make air flow from high to low (1). (iii) Day: sea breeze; night: land breeze (2).	2+1+2
	OR: (i) Troposphere, stratosphere, mesosphere, thermosphere, exosphere (2). (ii) Nitrogen 78%, oxygen 21%, others (1). (iii) Absorbs UV and traps heat (2).	2+1+2
20	(i) CFCs from refrigerators and aerosols (1). (ii) Skin cancer, eye damage, harm to ecosystems (2). (iii) The Montreal Protocol (1). (iv) The layer is slowly recovering (1).	1+2+1+1
	OR: (i) GHGs trap re-radiated heat (2). (ii) Rise in average temperature due to excess GHGs (1). (iii) Melting of glaciers, sea-level rise, extreme weather, monsoon changes (2).	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.